EG2300 Engineering Design II – Systems

Term Project Assignment

Design and Implementation of the EE8 Manufacturing Controller CPU

Industry-Style Digital Systems Design Brief

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2025/26

1 Objective

To design and implement a prototype **8-bit CPU** (EE8) in Logisim, capable of controlling a **batch diverter station** in a packaging/manufacturing line.

In the industrial scenario, your company has been contracted to deliver a low-cost embedded controller for a two-lane conveyor diverter. Items are detected by a sensor and must be routed to Gate A or Gate B according to a programmed batching policy. Reliability and safety interlocks are critical: the controller must behave predictably under operator stops, non-emergency faults, and illegal instruction conditions.

2 Project Context

You are part of a systems engineering team within an industrial automation supplier. A client operating a packaging and distribution plant requires a custom controller for a **batch diverter manufacturing station**.

Rather than using an off-the-shelf PLC, the client has commissioned a **custom CPU-based controller** to:

- Reduce unit cost for multiple identical lines,
- Improve transparency and auditability of the control logic,
- Enable future instruction-set extension for new station variants,
- Demonstrate robust safety behaviour suitable for industrial deployment.

Your implementation must comply with the provided **EE8 ISA specification (v4.0)** and demonstrate correct behaviour on the station I/O and platform safety logic.

3 Scope and Constraints

3.1 Minimum technical scope

Your CPU must implement:

- 8-bit datapath, 16-bit instruction word, and 4 registers (R0–R3).
- A 4-bit program counter addressing a 16-word ROM (addresses 0–15).
- Single-cycle fetch–decode–execute behaviour.
- Correct decode and execution of the EE8 instruction set.
- Correct platform safety behaviour (REBOOT, STOP, FAULT, SYS_ERR, CPU_HALT).
- One I/O interface option (A, B, or C), clearly documented and justified.

3.2 Systems engineering expectations

Your design should demonstrate:

- Modular structure (subcircuits), clear labelling, and maintainability.
- A clear datapath/control concept (even if implemented as combinational control).
- Evidence of trade-off decisions (e.g., I/O option selection, decode strategy).
- A risk-aware approach (what can go wrong, how you detect/limit impact).

4 Learning Outcomes (How You Will Be Assessed)

This project assesses your ability to:

- Apply digital design principles to develop a working system-level solution.
- Analyse and justify design decisions using engineering reasoning.
- Use simulation/modelling tools appropriately (Logisim) and recognise limitations.
- Consider safety, reliability, and quality in the context of an industrial system.
- Work effectively as a team and communicate design outcomes clearly.

5 Project Deliverables

5.1 1) Logisim Implementation (30%) – Group Mark

Submit a complete Logisim design that includes:

- CPU datapath and control (as appropriate for your architecture).
- Program memory and program loading mechanism (if required by the lab setup).
- Register file (R0–R3), ALU operations, instruction decode, and PC logic.
- I/O implementation (Option A, B, or C) and correct port behaviour.
- Platform safety behaviour and illegal-instruction response as specified.

Your Logisim implementation must include appropriate visual debugging and monitoring interfaces using LEDs and/or 7-segment displays.

These should be used, where appropriate, to display:

- Registers and flags,
- Program Counter (PC),
- Output port state (e.g., gate_sel, motor_en),
- Any additional status or extension signals implemented by your team.

Displays must be clearly labelled and arranged logically to support demonstration, debugging, and oral examination. The intent is that an external reviewer can observe system state transitions in real time during execution.

Marking criteria (Logisim):

Criterion	Marks
Correct ISA implementation (instructions execute as specified)	10
Correct PC behaviour and branching/jump logic	5
Correct I/O behaviour (chosen option) and signal semantics	5
Correct safety behaviour (STOP/FAULT/REBOOT/etc.)	5
Circuit organisation: modularity, labelling, neatness, visualisation	5
Total	**30**

5.2 2) Engineering Design Report (30%) – Group Mark

Maximum 2000 words.

Your report must be written as a professional engineering document. It must include:

- **Executive summary:** what you built and what it achieves.
- **Requirements and assumptions:** what "correct" means for this system.
- **Architecture:** block diagram(s) and design rationale.
- **Datapath and control design:** including truth tables, K-maps, state diagram and state tables, state/control logic, etc. as appropriate.
- **Safety and risk analysis:** hazards, failure modes, mitigations, and safety interlocks.
- **Verification and validation:** test plan, test cases, and evidence of correctness.
- **Limitations and future improvements:** including possible opcode extension (if relevant).

The report must be concise, technically focused, and free from unnecessary repetition. Explanations should be precise and directly relevant to the design decisions and analysis being presented. Marks will be deducted for repetitive content, redundant descriptions of the specification, or unnecessarily verbose explanations that do not add technical value.

Marking criteria (Report):

Criterion	Marks
Systems-level design explanation (clear structure and rationale)	8
Technical correctness and depth (datapath/control, decode, signals)	8
Risk and safety analysis quality (realistic, appropriate, justified)	5
Verification/testing quality (coverage, evidence, interpretation)	5
Clarity and professionalism (formatting, diagrams, readability)	4
Total	**30**

5.3 3) Presentation and Oral Examination (30%) – Individual Mark

Presentation length: 10 minutes (group presentation) followed by technical questioning.

Although you present as a group, **each student will receive an individual mark** based on:

- Technical understanding of the CPU and the station behaviour.
- Ability to explain design decisions and justify trade-offs.
- Ability to answer architecture-level questions (datapath, decode, control, safety).
- Understanding of risks, failure modes, and how your design behaves under faults.

Marking criteria (Presentation – individual):

Criterion	**Marks**
Clarity of explanation (structure, delivery, technical communication)	8
Depth of technical understanding (can explain *how* and *why*)	10
Defence of design choices (trade-offs, correctness arguments, limitations)	6
Professional communication (answers, terminology, confidence, precision)	6
Total	**30**

6 Assessment Summary

Component	**Weight**
Logisim Implementation (Group submission on Moodle)	30%
Engineering Design Report (**PDF** group submission on Moodle)	30%
Presentation + Oral Examination (Individual and in person)	30%
Total covered in this brief	**90%**

7 Guidelines

7.1 Teamwork and contribution

- Work as an engineering team: allocate roles (e.g., datapath lead, control/decode lead, verification lead, documentation lead).
- Ensure all members understand the full system, not only their subsystem.
- Clearly label subcircuits and signals; aim for a design that another engineer could maintain.

7.2 Testing and evidence

- Your report should include a clear test plan and evidence (screenshots, traces, tables of results).
- Include both nominal tests (correct programs) and robustness tests (fault, stop, reboot, illegal instruction behaviour, etc.).

7.3 Safety and reliability

- Treat the station as safety-relevant: the motor enable must behave correctly under halts/faults.
- Document failure modes (sensor bounce, missed pulses, illegal opcode, etc.) and how your design responds.

8 Failure Conditions (Non-exhaustive)

A submission may be considered inadequate if:

- The ISA is not fully implemented or does not match the specification.
- Safety behaviour is missing or incorrect (e.g., SYS_ERR not latching, HALT not freezing execution).
- The design cannot reliably execute and demonstrate the intended manufacturing control logic.
- During questioning, individual students cannot explain core parts of the design.

9 Final Note

This project is written in the style of an industrial design contract: correctness, robustness, and professional communication matter. Your goal is not only to build a working CPU, but to demonstrate that you can engineer a dependable system and justify it clearly.

I wish you a joyful teamwork!